

Q1. Short Questions & MCQ

- A. A leaf node of BST does not have an in-order predecessor! [1 Mark]
 a) True b) False
- B. A heap stored in an array is always sorted. [1 Mark]
 a) True b) False
- C. Which operation is used after deletion in a heap? [1 Mark]
 a) Heapify Up b) Heapify Down c) Traversal d) Sorting
- D. If a duplicate value is inserted into a max heap, will it still maintain the max heap property? [1 Mark]
 a) Yes b) No
- E. Given the following array(1 based indexing) representation of a binary tree, **draw** the corresponding tree. **Determine** whether it is a max heap. [2 Marks] Draw the tree here
 arr = [null, 50, 30, 40, 10, 60, 35, 25]
 This is a -
 a) Max Heap
 b) Not a Max Heap

Q2. Write a method named **kThSmallest** which will take the root of a Binary Search Tree and an integer k as parameters. The method should return the k-th smallest element present in the BST.

You may assume the following:

- $1 \leq k \leq$ total number of nodes in the tree
- All node values are unique
- The Node class is already defined with elem (Integer), left (Node), and right (Node)
- You do not need to implement the Node class

Constraints:

- You may create helper methods if needed
- You can not take any additional data structure except arrays.
- You are not allowed to use any instance or static variables

Sample Input1	Output1	Sample Input2	Output2
<pre> 5 /\ 3 6 /\ 2 4 </pre> k = 3	4	<pre> 8 /\ 3 10 /\ \ 1 6 14 /\ / 4 7 13 </pre> k = 5	7
<u>Explanation1:</u> The in-order traversal of the BST gives: 2, 3, 4, 5, 6. The 3rd smallest element is 4.		<u>Explanation2:</u> The in-order traversal of the BST gives: 1, 3, 4, 6, 7, 8, 10, 13, 14 The 5th smallest element is 7.	

Complete this method in your preferred language:

JAVA	PYTHON
<pre> public static int kThSmallest(TreeNode root, int k){ //ToDo </pre>	<pre> def kThSmallest(root, k): #ToDo </pre>